TEACHER VERSION. Do not hand out. Same worksheet as the pupil copy with the **answers in red**. Pupils get `Guessing-Pi-Worksheet.pdf` (or the `.docx`). Slides: `Slides/decks/monte-carlo-pi`. Code answer: `code/pi_darts_solution.py`.

Guessing π: independent worksheet

A self-contained version of the 15 minute lesson, for pupils working alone or at home. It follows the same order as the slides. **Prior learning:** variables, `if` / `else`, `for` loops and `%` (Intro to Python). **Equipment:** Spyder, a calculator, a pencil. Python's built-in `random` module only.

Part	Topic	Minutes
Part 1	The question	3
Part 2	What is π? (maths first)	6
Part 3	Random numbers in Python	5
Part 4	Throw one dart	4
Part 5	Is the dart inside? (by hand)	5
Part 6	Inside or outside in code	5
Part 7	Count 10,000 darts	6
Part 8	The trick	4
Part 9	More darts, better guess	5
Part 10	Monte Carlo, stretch, checklist	2

About 45 minutes in total. Parts 4 to 7 are the coding core. Pupils who are quick can do the two stretch tasks and the million-dart run.

Watch for: forgetting the colon after `if` / `for`; missing indents; `x^2` instead of `x*x`; putting `inside = 0` inside the loop (the count resets every dart); `inside = inside + 1` not indented under the `if` (counts every dart); a final `print` indented inside the loop; typing `1,000,000` with commas (use `1000000`).

Numbers to expect: 10,000 darts gives about 7,850 inside (a spread of roughly 7,800 to 7,900) and a guess within about 0.02 of 3.14159. One million darts takes a few seconds and is usually within 0.005.

Guessing π with Random Darts

Work through this on your own. You will find the value of π using nothing but random numbers and Python. About 45 minutes.

Name: _____ Class: _____

You need: Spyder (or any Python editor). Type your code into a new file and press Run. You only need `import random`. Work through the Parts in order and tick them off at the end.

Part 1 The question

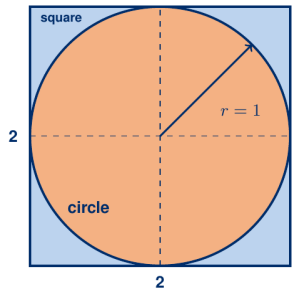
1 How could you find out the value of π **without** already knowing it? Write down one idea.

Any sensible idea, e.g. measure round a circle with string and divide by the width across. Today's method uses random darts instead.

Today you will do it with a dartboard, random numbers and Python. $\pi \approx 3.14159\ldots$ and its digits never end, so any method only gives an **estimate**.

Part 2 What is π? The maths first

π is in every circle's area: **area =** $\pi \times$ **r²**. Here is a circle with radius 1 inside a square that is 2 wide and 2 tall.



2 Area of the circle (r = 1): $\pi \times 1^2 =$ **π** Area of the square: $2 \times 2 =$ **4**

3 Imagine throwing darts at random so they land evenly over the square. The fraction that land inside the circle is circle area ÷ square area = **π** ÷ **4**

4 If $\pi \div 4 \approx 0.785$, about how many out of 1000 darts land inside the circle? **about 785**

The four quarters of the picture are identical, so we only need **one**. Our dartboard is the top-right quarter: a square 1 wide and 1 tall, with a quarter circle of radius 1 whose centre is the bottom-left corner.

5 Area of the 1 by 1 square: **1** Area of the quarter circle: $\frac{1}{4} \times \pi \times 1^2 =$ **π ÷ 4**. So the fraction inside is still $\pi \div 4$.

Part 3 Random numbers in Python

Type these two lines into the console (bottom right in Spyder), one at a time:

```
import random
random.random()
```

6 Run `random.random()` three times. Write the numbers you get: **any number** **0 to 1** **all different**

7 Every number you got is between **0** and **1**.

`random.random()` is **uniform**: every value from 0 to 1 is equally likely. So if we use one random number for x and another for y, the dart lands evenly anywhere in the square.

8 Why would our π come out wrong if the darts bunched up in one corner? Write one sentence.

The number of darts inside would no longer match the area of the quarter circle, so inside ÷ total would not equal $\pi \div 4$.

Part 4 Throw one dart

Type this into a new file and run it:

```
import random

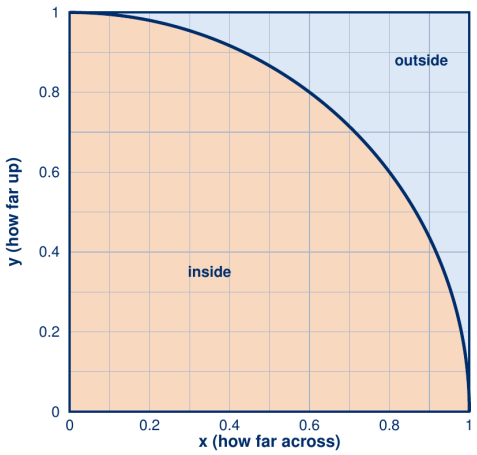
x = random.random()  # how far across
y = random.random()  # how far up

print(x, y)
```

9 Run it three times. Write each x and y in the table, then mark each dart with a cross on the dartboard.

Dart	x	y
1		
2		
3		

Any values between 0 and 1. Check the crosses are plotted at (x, y), reading x along the bottom and y up the side.



Part 5 Is the dart inside?

A dart is **inside** if its distance from the bottom-left corner is at most r = 1. By Pythagoras, the distance is $\sqrt{x^2 + y^2}$. Fill in the steps:

Step	Maths
inside means	distance $\leq$ r
square both sides	$x^2 + y^2 \leq$ **r^2**
r = 1, and 1^2 = **1**	$x^2 + y^2 \leq$ **1**
in Python	`x*x + y*y <=` **1**

In Python, `x^2` does **not** square x. Write `x*x` (or `x**2`).

10 Work out each dart by hand. Dart 1 is done for you. Use the rule: inside if x*x + y*y is 1 or less.

Dart	x	y	x*x + y*y	Inside or outside?
1	0.6	0.5	0.61	inside
2	0.9	0.8	**1.45**	**outside**
3	0.3	0.7	**0.58**	**inside**
4	0.2	0.95	**0.9425**	**inside**
5	0.95	0.4	**1.0625**	**outside**

Part 6 Your turn: inside or outside in code

Fill in the two blanks, then type it in and run it **five** times.

```
import random

x = random.random()
y = random.random()

if x*x + y*y <= 1 :
    print("inside!")
else:
    print("outside ")
```

11 Out of your 5 runs, how many said inside? **usually 3 or 4** About how many would you expect? **about 4 (3 in every 4)**

Hints (printed upside down for pupils)

Stretch: use a `for` loop to throw 10 darts. Hint 1: the first line is `for i in range(10):`. Hint 2: everything you want repeated is indented underneath it.

Part 7 Count 10,000 darts

Now throw lots of darts and **count** the ones that land inside. `inside = inside + 1` adds one to the count. Fill in the three blanks and run it three times.

```
import random

total = 10000
inside = 0

for i in range(total):
    x = random.random()
    y = random.random()
    if x*x + y*y <= 1:
        inside = inside + 1

print(inside)
```

12 Write the three numbers it printed: **close to 7,850** **close to 7,850** **close to 7,850**

13 Divide your first number by 10,000: **about 0.785**. Now use a calculator for $\pi \div 4$ (use 3.14159): **0.7854**. Are they close?

Part 8 The trick

We know that **inside** $\div$ **total** $\approx \pi \div$ **4**. Multiply both sides by 4:

14 $\pi \approx$ **4** $\times$ **inside** $\div$ **total**

Add these two lines at the **end** of your program (not indented):

```
pi_guess = 4 * inside / total
print(pi_guess)
```

15 Run it three times and fill in the table. For the last column, work out how far each guess is from 3.14159.

Run	My guess for π	How far from 3.14159?
1		
2		
3		

Guesses are usually within about 0.02 of 3.14159. We found π without ever typing 3.14.

Part 9 More darts, better guess

16 Change **only** the number in `total = ...` and fill in the table. One million darts takes a few seconds.

total	Guess 1	Guess 2	Guess 3
100	**3.12**	**2.88**	**3.44**
10,000	**3.1588**	**3.1528**	**3.1352**
1,000,000	**3.1424**	**3.1432**	**3.1436**

Example answers only: pupils' numbers will differ. Look for guesses that vary a lot at 100 and barely vary at 1,000,000.

17 What do you notice as the number of darts gets bigger? Write one or two sentences.

The guesses get closer together and closer to 3.14159. With few darts luck matters a lot; with many darts it evens out.

18 Why can this method never give π exactly?

It depends on random darts, so it is always an estimate. (π's digits never end either.)

Part 10 This has a name

A **Monte Carlo method** uses lots of random tries to answer a question. It is named after a famous casino in Monaco. People use it for weather forecasts, video game AI and physics simulations.

19 Think of something else that is too hard to work out exactly, but that you could estimate by trying it thousands of times at random. Write it down.

Open answer, e.g. the chance of winning a game, how long a queue gets, the weather.

Hints (printed upside down for pupils)

Stretch: print the guess every 1,000 darts so you can watch it settle. Hint 1: do this inside the loop. Hint 2: `i % 1000 == 0` is true on every 1,000th dart. Hint 3: `if i % 1000 == 0:` then, indented, `print(4 * inside / (i + 1))`.

Finished What I can do now

- ☐ I know `random.random()` is uniform, so darts land evenly
- ☐ I know the distance from the corner is $\sqrt{x^2 + y^2}$
- ☐ I know inside means `x*x + y*y <= 1`
- ☐ I know the quarter circle is $\pi \div 4$ of the square
- ☐ I know $\pi \approx 4 \times$ inside $\div$ total